Early Fake News Detection: datasets and exploratory analysis

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1. Chosen datasets

We classify a news item or claim as fake or real from only the *first part* of its spread on Twitter (first h hours or first k engagements), which requires propagation trees with a timestamp on every node. We use four datasets from two independent sources (Table 1). **PolitiFact** and **GossipCop** come from UPFD [1], the dataset of our main paper (politics and celebrity news; per-node timestamps from the UPFD authors' repository). **Twitter15/16** [2] differ from the main paper: a different collection and domain (rumour claims), four veracity classes, larger and much faster cascades, and no user features. They let us test whether conclusions transfer beyond UPFD. For binary experiments we map *false*→fake, *true*+*non-rumor*→real and drop *unverified* (veracity unknown), which makes Twitter ∼2:1 real:fake.

Table 1: Datasets after preprocessing. Labels: real/fake (UPFD), non-rumor/false/true/unverified (Twitter). Span = median time from first to last engagement.

Dataset	Source	Cascades	Labels	Mean nodes	Span	Node	Node features
PolitiFact	UPFD [1]	314	157 / 157	131	219 d	post	user history (BERT, spaCy), profile
GossipCop	UPFD [1]	5464	2732 / 2732	58	9 d	post	as above
Twitter15	Ma et al. [2]	1490	374 / 370 / 372 / 374	402	8 d	user	none (source text only)
Twitter16	Ma et al. [2]	818	205 / 205 / 207 / 201	425	8 d	user	none (source text only)

2. Data model and preprocessing

All datasets are converted (`src/build_data.py`; analysis in `notebooks/eda.ipynb`) into one format: a rooted tree per cascade (root = news or source tweet, edge = parent → child propagation) with nodes sorted by time, so that parent time ≤ child time. Hence every early window is a prefix of the node list and always a connected tree, and the same window can be given to a model either as a *graph* or as a time-ordered *sequence* (Fig. 1), which is exactly the comparison our project makes. Raw Twitter trees are not strict trees: ∼1% of nodes have several candidate parents, which causes 98% of time-inverted edges. We keep the most recent earlier parent, drop self-loops, and clamp 736 posts that predate the source tweet to t=0. Every fix is a column or flag (raw data untouched). We also flag leakage sources: 169 cascades occur in both Twitter15 and Twitter16, some source texts repeat, and UPFD contains 39 groups of different news items with identical cascades.

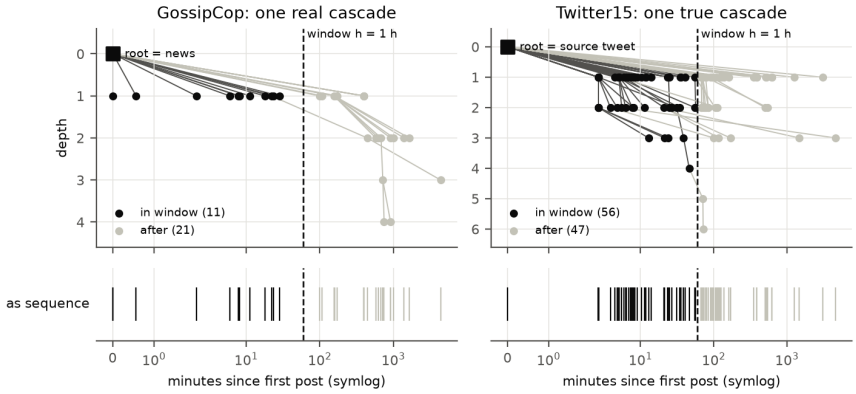


Figure 1: Two real cascades as trees (x: time, y: depth) and as sequences. Left of the dashed line: the input of a 1-hour early detector.

3. Basic analysis: size and shape

Cascade size depends on the class, but not in the same direction everywhere (Fig. 2): fake cascades are smaller in GossipCop (median 22 vs 70 nodes) and Twitter (∼210 vs ∼350), but larger in PolitiFact (104 vs 58). Trees are shallow and star-like: the median share of nodes attached directly to the root is 0.6–0.9 and 70% of GossipCop cascades have depth 2; the depth distributions of the two classes overlap strongly.

4. Specific analysis: what can be seen early?

Growth speed differs by orders of magnitude (Fig. 3). After 1 h a Twitter cascade has ∼60% of its final size (median ∼160 engagements), GossipCop 25% and PolitiFact 5% (median 2 engagements); real PolitiFact cascades keep growing for years. A fixed "first h hours" is therefore not comparable across datasets, so we will use per-dataset time grids together with first-k windows.

Simple activity features are strong early, structure is not (Fig. 4). The number of engagements in the window alone reaches ROC AUC 0.89 on GossipCop after 6 h (0.79 on full cascades) and 0.81 on Twitter16 after only 10 minutes (0.66 on full cascades): early activity is *more* informative than the whole cascade. Tree shape (maximum depth, share of root children) stays at AUC 0.5–0.6 in most windows, because early trees are mostly stars. PolitiFact separates only after several days.

Fake content starts slower here (Fig. 5). The 5th engagement arrives after a median of 8 min for real and ∼5 h for fake GossipCop news (AUC 0.85 from this one number), and after 0.6–0.7 vs 1.5–1.7 min on Twitter. This contradicts the common "fake spreads faster" finding [3] at the start of a cascade; a likely reason is that real stories come from large

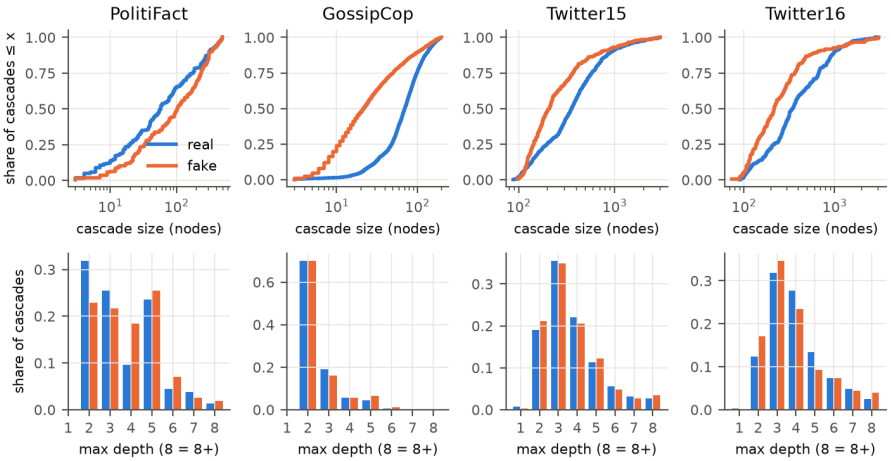


Figure 2: Top: cumulative distribution of cascade size; bottom: distribution of maximum depth (binary labels).

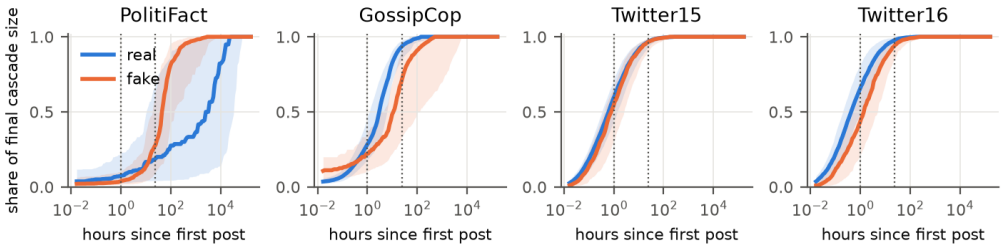


Figure 3: Share of the final cascade size observed t hours after the first post (median and interquartile range per class).

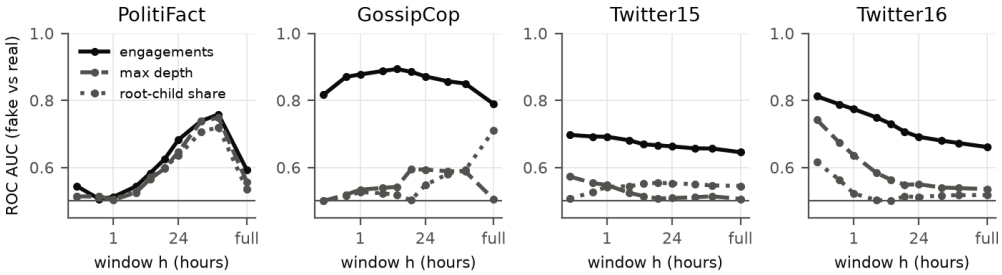


Figure 4: Separability (ROC AUC, direction-free) of single features computed on the first h hours; "full" = whole cascade.

mainstream accounts. PolitiFact behaves differently (fake becomes faster after some hours).

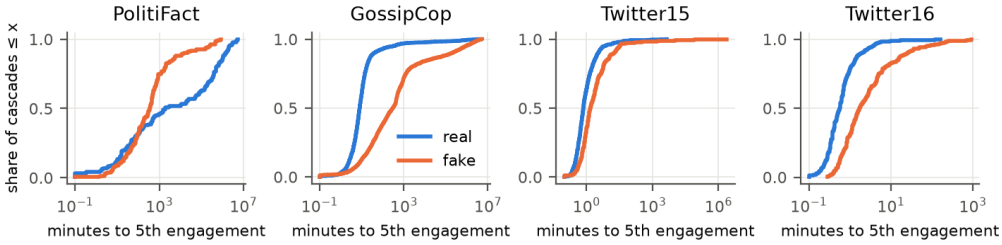


Figure 5: Cumulative distribution of the time until the 5th engagement.

Users and splits. 10–21% of users take part in several cascades, and 60–82% of them only ever appear in one class, against 30–47% after shuffling the labels. User identity is thus a strong signal (it explains why UPFD's user-history features work), but also a leakage channel. The official PyG split of UPFD is small (62 training cascades in PolitiFact) and puts 18 of the 39 duplicate-cascade groups on both sides of the split.

References. [1] Y. Dou et al., User Preference-aware Fake News Detection, SIGIR 2021. [2] J. Ma, W. Gao, K.-F. Wong, Detect Rumors in Microblog Posts Using Propagation Structure via Kernel Learning, ACL 2017. [3] S. Vosoughi, D. Roy, S. Aral, The spread of true and false news online, Science 359, 2018.